

Draft - expert review pending

This unit is complete and passes the platform's automated checks, but no reviewer has read it yet. Treat its sources and claims with care.

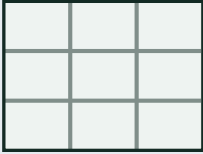
Topic 4.2 - Perimeter, area, volume and basic shapes

A real classroom situation

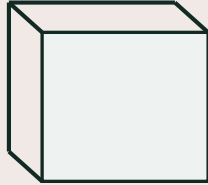
One corner, three questions



Perimeter
the boundary, one-dimensional,
measured in metres
"how much skirting?"



Area
the cover, two-dimensional,
measured in square metres
"how many tiles?"



Volume
the fill, three-dimensional,
in cubic metres or litres
"how much water?"

The same corner answers three different questions;
confusing them is expensive: skirting is not bought by the square metre.

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The headteacher of a primary school in Matiari has one corner of a classroom and three questions. Question one: how much skirting board does the corner need? Question two: how many floor tiles cover it? Question three: how much water fits in the tank that will sit there? Same corner, three different quantities: the boundary (skirting), the cover (tiles), the fill (water). A carpenter, a mason and a plumber would each compute something different, and confusing them is

expensive - order skirting board by square metres and you will drown in timber. This topic gives the three quantities their names, their formulas and their units, so the right one answers the right question.

Explanation

Perimeter, area, and volume

Perimeter is the distance around a shape: the fence line. Area is the amount of surface a shape covers: the carpet. Volume is the amount of space a solid fills: the water in the tank. The diagram above shows all three on one classroom corner; the units keep them apart - metres (m) for perimeter, square metres (m²) for area, cubic metres (m³) or litres for volume.

Quantity	Formula	Worked classroom example
Rectangle perimeter	$2 \times (\text{length} + \text{width})$	a classroom 8 m by 6 m needs 28 m of skirting
Rectangle area	$\text{length} \times \text{width}$	the same classroom has 48 square metres of floor
Triangle area	$(\text{base} \times \text{height}) / 2$	base 1.2 m, height 0.8 m: 0.48 square metres
Circle circumference	$2 \times \pi \times \text{radius}$	radius 4 m: 25.12 m of border
Circle area	$\pi \times \text{radius squared}$	radius 4 m: 50.24 square metres
Box volume	$\text{length} \times \text{width} \times \text{height}$	2 m by 1 m by 1.5 m holds 3 cubic metres

pi is about 3.14; 1 cubic metre is 1,000 litres.
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The table above collects the formulas with a worked classroom example for each. Three deserve a closer look.

Rectangle perimeter and area. A classroom 8 metres by 6 metres: perimeter = $2 \times (8 + 6) = 28$ metres of skirting; area = $8 \times 6 = 48$ square metres of tiles. Notice the perimeter counts the boundary once around, while the area counts

the unit squares inside - the same two measurements serve both, but they answer different questions.

Triangle area. A triangular notice board of base 1.2 metres and height 0.8 metres: $\text{area} = (1.2 \times 0.8) / 2 = 0.48$ square metres. Half of the enclosing rectangle: that is where the divide-by-two comes from, and picturing it prevents the most common triangle error (forgetting the half).

Box volume. A store box 2 m by 1 m by 1.5 m: $\text{volume} = 2 \times 1 \times 1.5 = 3$ cubic metres. Because 1 cubic metre is 1,000 litres, the box holds 3,000 litres - the conversion from Topic 4.1 riding along.

Basic geometric shapes and properties

A rectangle has four right angles and opposite sides equal; a square is a rectangle with all sides equal; a triangle has three sides whose angles sum to 180 degrees; a circle is every point at one fixed distance (the radius) from a centre. The circle's own numbers are special: its boundary (the circumference) is about 6.28 radii around - the number pi (about 3.14) times the diameter - and its area is pi times the radius squared. For school work, taking pi as 3.14 is standard.

A common misconception to correct. Many trainees believe "a bigger area always means a bigger perimeter". The two can move separately: a long thin corridor of 2 metres by 20 metres has area 40 square metres and perimeter 44 metres, while a near-square room of 6 by 7 metres has more area (42) with less perimeter (26). Fencing buys perimeter; carpet buys area; a school that confuses them buys the wrong one.

Activity: Fence or carpet

Work in groups of three. Your tutor gives each group four room plans on paper (a classroom, a corridor, a store room and an odd-shaped staff room made of two rectangles). For each plan, compute the perimeter and the area, then decide: if the school can afford either 100 metres of skirting or 700 square metres of tiles in total, which rooms get which first? Defend the priority order in two sentences. About 25 minutes.

Check your understanding

1. Compute the perimeter and area of a classroom 9 metres by 5 metres.
(*Apply*)
2. A triangular flower bed has base 3 metres and height 2 metres. Compute its area, and explain in one sentence where the divide-by-two comes from.
(*Apply*)
3. A water tank is a box 1.5 m by 1 m by 1 m. Compute its volume in cubic metres and in litres. (*Apply*)
4. A circular flower bed has radius 4 metres. Compute its circumference and area (take pi as 3.14). (*Apply*)
5. A corridor of 2 m by 20 m and a room of 6 m by 7 m: state which has the larger area and which the larger perimeter, and say what a school buys with each quantity. (*Analyze*)

Summary

Perimeter is the boundary in metres, area the cover in square metres, volume the fill in cubic metres or litres; the units are part of the answer. The rectangle, triangle, circle and box formulas cover most school measurement, and each formula's shape (the half in the triangle, the pi in the circle) is a picture, not a spell. Area and perimeter move independently: fencing buys one, carpet the other, and the school that confuses them buys wrongly.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can compute perimeter, area and volume for rectangles, triangles, circles and boxes, with units attached.
- ☐ I can explain where each formula's key step comes from (the half, the pi).
- ☐ I can state, for a given school task, whether it buys perimeter, area or volume.

Try this at your practicum school

With permission, measure one rectangular room's length and width (pace it if no tape is available, and note that your method is an estimate). Compute its perimeter and area, then ask the caretaker how much skirting or tiling the room actually used, and compare.

Summative task

The school wants to fence a rectangular kitchen garden 12 metres by 8 metres, carpet a staff room 5 m by 4 m, and install a cylindrical water tank of radius 1 metre and height 2 metres. In a short written answer: (a) compute the fencing needed (perimeter), the carpet needed (area), and the tank's volume in litres (pi as 3.14), (b) state the unit of each answer, and (c) check one of the three answers with an estimation sentence, and write the whole answer so a colleague could re-check every step.

Mini-rubric (each criterion 0-2 marks):

- *Fencing*: perimeter = $2 \times (12 + 8) = 40$ metres.
- *Carpet*: area = $5 \times 4 = 20$ square metres.
- *Tank*: volume = $3.14 \times 1 \times 1 \times 2 = 6.28$ cubic metres = 6,280 litres, with the unit stated on every answer.
- *Check and clarity*: one estimation check shown (for example, the tank is roughly a 2 m by 2 m by 2 m box = 8,000 litres, so 6,280 is plausible) and the steps are laid out for re-checking. **(Analyze)**

Further reading

- Marecek, L., Anthony-Smith, M., & Mathis, M. H. (2020). *Prealgebra 2e* (Ch. 9 Math Models and Geometry: linear, square and cubic measures; rectangles, triangles and trapezoids; volume and surface area). OpenStax.
<https://openstax.org/details/books/prealgebra-2e>
- National Curriculum Council, Ministry of Federal Education and Professional Training, Government of Pakistan. *NCP Mathematics Progression Grid (Grades 1-12)*.

<https://ncc.gov.pk/Detail/ZjgzYzg2MmMtZDc0Zi00NjEzLTk5ZmYtZGJiNjc5ODljOGUx>

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